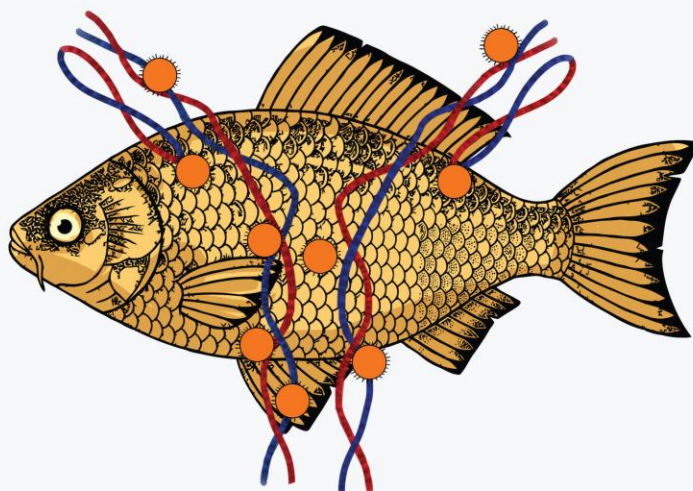


IMMUNOLOGICAL TECHNIQUES USING FISH MODEL

A Laboratory Manual



R. Dinakaran Michael

A Manual for a Laboratory Course on Immunology
(in Life Science Programmes at Bachelors and Masters Levels on
Biotechnology, Microbiology, Biology, Zoology, Veterinary and Fisheries Sciences and
related subjects in Life Sciences)

Prof. Dr. R. Dinakaran Michael, PhD
Dean of Life Sciences and
Director, Centre for Fish Immunology
Vels Institute of Science, Technology and Advanced Studies,
Pallavaram, Chennai – 600117, India
dean.sls@velsuniv.ac.in

About the Manual

Hello there, immunology teachers of colleges and universities! Are you still struggling to design, develop and conduct laboratory exercises to demonstrate concepts of immunology to your undergrad and postgrad students using rabbits, guinea pigs and mice without an approved animal house and animals? All your problems are over. Here is an innovative and 'easy to follow' laboratory manual which has the procedures of many immunological techniques and assays using fish model. Using fish model is to avoid the constraints of i) having and maintaining spacious, hygienic, air-conditioned approved animal house ii) procuring and maintaining mammalian models like rabbits, guinea pigs and mice which are traditionally used for immunology laboratory exercises and iii) stringent animal ethical guidelines. Fishes do have most of the immunological components, mechanisms and molecules found in higher vertebrates like rabbits, guinea pigs and mice though in a simpler style. Hence, fish is as an ideal vertebrate model for teaching and demonstrating the essential immunological concepts.

The immunological exercises covered in this manual includes but not restricted to the basic techniques like dissection of lymphoid organs, preparation of leucocytes from these organs, antigen preparation, immunization and bleeding technique, and serum separation and assays like non-specific immune mechanisms like lysozyme activity and production of reactive oxygen by phagocytes, specific immune responses like antibody production, allograft rejection and DTH reaction, expression of immune related genes and vaccination and vaccine efficacy testing. Many of these techniques are being routinely used in institutions where the author served and also in institutions whose faculty fellows underwent training in the large number of training workshops conducted by the author.

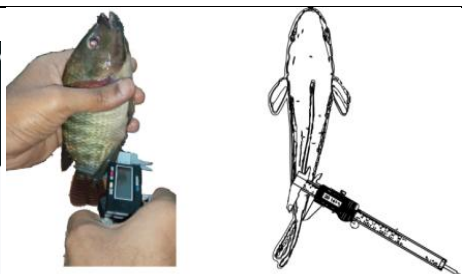
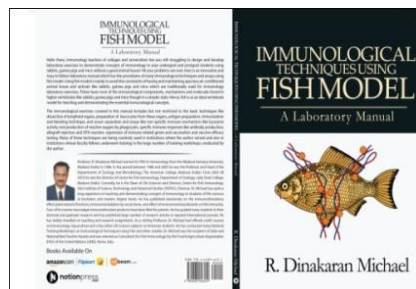
About the Author

Professor R. Dinakaran Michael earned his PhD in Immunology from the Madurai Kamaraj University, Madurai (India) in 1986. In the period between 1986 and 2003 he was the Professor and Head of the Departments of Zoology and Microbiology at The American College, Madurai (India). From 2003 till 2010, he was the Director of Centre for Fish Immunology, Department of Zoology, Lady Doak College, Madurai (India). Currently, he is the Dean of Life Sciences and Director, Centre for Fish Immunology, Vels Institute of Science, Technology and Advanced Studies (VISTAS), Pallavaram, Chennai. Dr. Michael has quite a long experience in teaching and demonstrating concepts of Immunology to students of life sciences at bachelors and masters degree levels. He has published extensively on the immunostimulatory effect plant extracts/fractions, immunomodulation by social stress, and effect of environmental pollutants on fish immunity. Four of his marine macroalgal immunostimulant products have been filed for patents and are awaiting awards. He has guided many students in their graduate and doctoral research and has published large number of research articles in reputed international journals. He has widely travelled on teaching and research assignments. As a visiting Professor, Dr. Michael had offered credit courses on Immunology, Aquaculture and a few other Biology subjects to college/university students in USA. He has conducted many National Training Workshops on Immunological Techniques Using Fish and other models. Dr. Michael was the recipient of State and National Best Teacher Awards and was selected as Consultant (for Fish Immunology) by the Food & Agriculture Organization (FAO) of the United Nations (UNO), Rome, Italy.

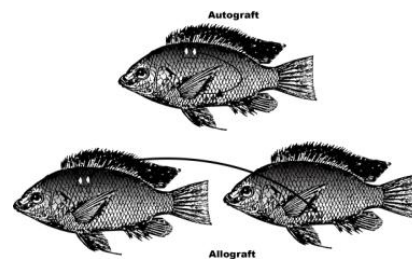
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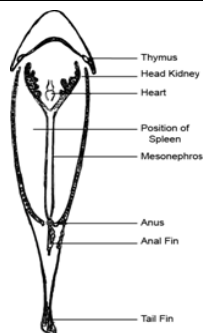
Some of the Figures in the Manual



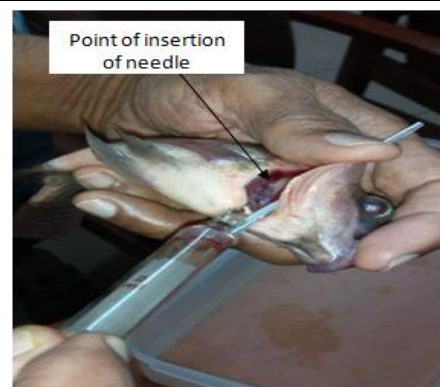
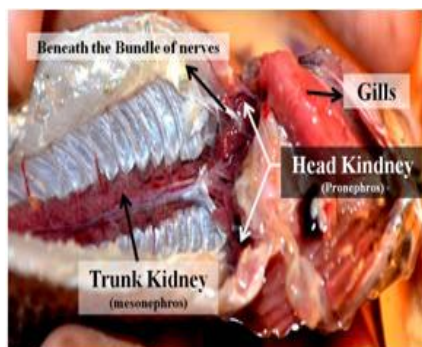
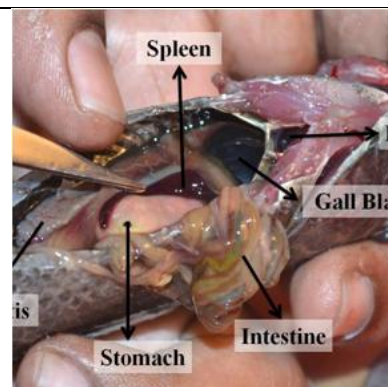
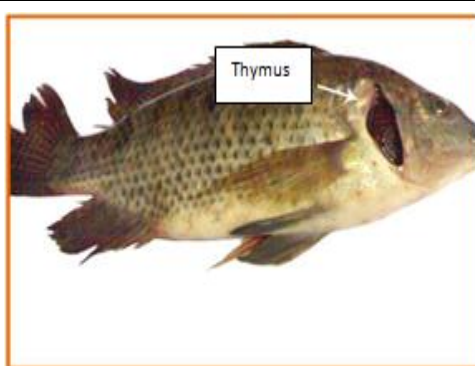
DTH: Measurement of Fish caudal peduncle thickening (equivalent of mouse foot pad thickening) using vernier.



Scale Autograft and Allograft



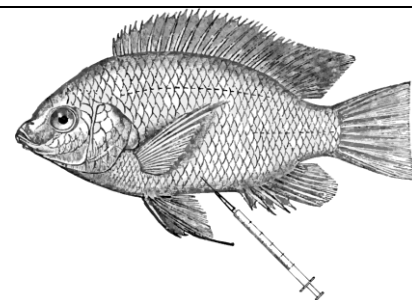
Fish Lymphoid Organs



Fish tubs with aerators



Amplified PCR products of TNF- α and IFN- γ with markers



Intraperitoneal injection

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